E-Commerce Sales Data Analysis

WeIntern Pvt. Ltd. — Week 1 Data Science Internship Assignment

1. Objective

The objective of this assignment is to clean an e-commerce sales dataset, analyze sales and customer patterns, calculate key performance indicators, and present findings through data visualizations.

2. Dataset Overview

Metric	Value
Dataset	E-commerce Sales Dataset
Total Records	10,000
Original Columns	10
Missing Values	0
Duplicate Rows	0
Invalid Dates	0
Date Range	2023-01-01 to 2023-12-31

3. Data Cleaning Summary

The dataset was checked for missing values, exact duplicate rows, repeated order IDs, invalid dates, negative or zero quantities, invalid prices, inconsistent labels, and mismatches between quantity multiplied by unit price and total price. The completed validation checks reported zero missing values, zero exact duplicate rows, zero invalid dates, zero negative quantities, zero zero-priced units, and zero price mismatches. The original raw dataset was retained separately from the cleaned dataset.

4. KPI Summary

KPI	Value
Total Recorded Order Value	7,627,241.97
Total Orders	10,000
Average Order Value (AOV)	762.72
Total Units Sold	30,097
Total Unique Customers	10,000
Top Revenue Category	Sports & Outdoors
Top Revenue-Generating Product	fae976be-6f36-43e3-b046-6e2d59357535

5. Sales Trend Analysis

The highest recorded category revenue was from Sports & Outdoors (1,313,735.43). The lowest was from Beauty & Health (1,201,765.08). The month with the highest recorded sales was 2023-12 (708,594.32), while the lowest was 2023-11 (552,576.79).

6. Customer Insights

The dataset contains 10,000 unique customers. Customer-level analysis was performed to compare recorded order values, unique order counts, and units associated with each customer. The highest recorded spending

customer was cd1472e9-2a12-484d-b496-fc6638dc948d, with a recorded value of 2,499.78.

7. Visualizations

The analysis includes a category-wise revenue bar chart, monthly sales trend line chart, order status pie chart, payment type bar chart, weekly sales trend, and a combined sales dashboard. These visuals support comparisons across categories, time periods, order statuses, and payment methods.

8. Business Findings

1. Sports & Outdoors had the highest recorded category revenue.
2. Beauty & Health had the lowest recorded category revenue.
3. 2023-12 had the highest recorded monthly sales.
4. 2023-11 had the lowest recorded monthly sales.
5. Pending was the most common order status.
6. PayPal was the most frequently used payment type.

9. Recommendations

1. Investigate successful sales strategies used by the highest-revenue category.
2. Review the lowest-revenue category to identify opportunities for improvement.
3. Study monthly sales patterns to support inventory and promotional planning.
4. Monitor Pending, Cancelled, and Refunded orders to identify possible operational issues.
5. Review payment-method usage when planning customer payment options.
6. Use customer purchasing patterns to design suitable retention and loyalty campaigns.

10. Limitations and Assumptions

Revenue and average order value in this report are calculated using the total_price values for all recorded order statuses, including Completed, Pending, Cancelled, and Refunded. Therefore, these figures represent recorded order value and should not be interpreted as completed-sales revenue. The dataset does not establish the reasons behind observed patterns; further investigation would be needed before making causal conclusions.